

1 Arithmetic symmetric frontier cost and determinant growth

This section proves finite restrictions on the spectral positions of the original arithmetic packet. It retains the actual theta source, the measure $|2\xi/h|^2 dy/(2\pi)$, all zero multiplicities, and the unscaled orbit coordinates. No purity assertion is used. Stars denote conjugate transpose in the displayed finite coefficient bases, and Hilbert adjoint for maps into the source Hilbert space. For Hermitian matrices, $U \preceq V$ means that $V - U$ is positive semidefinite. The one-factor quotient coefficient basis is always $[1]_h, [t]_h, \dots, [t^{d-1}]_h$, with its specified tensor and orbit inclusions below.

Authoritative source pins. The retained construction is in the original messages A1829 and A1859 of `audit_segment_U0041_U0054.md`. The line numbers here are literal lines in that source, whose SHA-256 is recorded in the accompanying audit. A1829, lines 4956–5071, specifies the source, the arithmetic function, the full-jet map, and its exact polynomial kernel; lines 5073–5169 give the canonical minimum and its metric; lines 5383–5561 give the joint tensor source and actual next relation layer. A1859, lines 5693–5874, gives the unscaled polynomial coordinates, the relation isomorphism, and the exact metric update; lines 5876–6040 give the recurrence and finite lowering bound; lines 6042–6197 give the signed symmetric summand and its unscaled orbit factors. The aggregate inequalities proved below are new consequences of those retained objects.

1.1 The retained arithmetic source and its canonical finite metric

Keep the original operator, theta map, and arithmetic function

$$\begin{aligned} D &= -x\partial_x, & \Theta\phi(x) &= \sum_{m \neq 0} \phi(mx), \\ \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & f_0 &= \Theta\phi_*, \\ g(s) &= \mathcal{M}f_0(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \end{aligned} \tag{FD.1}$$

Here $\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1}dx$, on its original source domain. Choose a nonempty finite packet of actual zeros and retain the full order m_ρ at every

selected centre:

$$\begin{aligned} h(t) &= \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \\ E_h &= \mathbb{C}[t]/(h), \quad v_h = [g/h]_h \in E_h^\times. \end{aligned} \tag{FD.2}$$

The bracket $[\cdot]_h$ means the complete local Taylor jets modulo h ; it includes every nilpotent jet at a multiple zero. Since each selected order is the full order of that zero, g/h is nonzero at every selected centre, proving that v_h is a unit in the finite quotient. Let A_h be multiplication by $[t]_h$ on E_h .

The retained actual source inverse supplies F_h with

$$h(D)F_h = f_0, \quad \mathcal{M}F_h = g/h, \quad \mathcal{T}_h(P) = P(D)F_h. \tag{FD.3}$$

Its exact identities, retained from the original source construction, are

$$\mathcal{M}\mathcal{T}_h(P) = (g/h)P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h\mathcal{T}_h(P) = v_h[P]_h. \tag{FD.4}$$

Here $J_h F = [\mathcal{M}F]_h$ is the full Mellin-jet map on the original analytic source. The second identity gives an actual theta primitive, and the last identity includes the arithmetic coefficient v_h . The map $\mathcal{T}_h$ is linear and injective: a zero image has identically zero Mellin transform $(g/h)P$, which forces $P = 0$.

The Hilbert observation remains $\mathcal{H} = L^2((0, \infty), dx)$. The exact change of variables and Fourier map

$$F \mapsto (y \mapsto e^{y/2} F(e^y)) \mapsto \left(t \mapsto \int_{\mathbb{R}} e^{y/2} F(e^y) e^{ity} dy \right)$$

have target measure $dt/(2\pi)$ and composite $F \mapsto \mathcal{M}F(1/2 + it)$. Consequently the arithmetic polynomial measure is exactly

$$d\nu_h(y) = \left| \frac{g(\frac{1}{2} + iy)}{h(\frac{1}{2} + iy)} \right|^2 \frac{dy}{2\pi}. \tag{FD.5}$$

At selected zeros on the integration line, the quotient is evaluated as the entire function g/h . The finiteness of every polynomial moment can also be checked directly from the retained factors. If $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$, its alternating coefficient partial sums have absolute value at most one. Partial summation gives $|\eta(s)| \leq |s|/\Re s$ for $\Re s > 0$. The identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$, first obtained by splitting the absolutely convergent series and then continued analytically, implies

$$|\zeta(\frac{1}{2} + iy)| \leq \frac{2|\frac{1}{2} + iy|}{\sqrt{2} - 1},$$

because $|1 - 2^{1/2-iy}| \geq \sqrt{2} - 1$. Fix $0 < \theta < \pi/2$. Rotating the Euler gamma integral through angle $\text{sgn}(y)\theta$ gives

$$|\Gamma(\frac{1}{4} + iy/2)| \leq \Gamma(\frac{1}{4})(\cos \theta)^{-1/4} e^{-\theta|y|/2}.$$

For $y = 0$, the same bound follows from the unrotated integral. The rotation follows by integrating $z^{1/4+iy/2-1}e^{-z}$ around a truncated sector using its continuous logarithm there. The small circular arc tends to zero as its radius to the power $1/4$; the large arc tends to zero because $\cos \theta > 0$. Taking absolute values on the rotated ray yields the displayed bound. Retaining the factors in (FD.1) therefore gives

$$|g(\frac{1}{2} + iy)|^2 \leq \frac{4\pi^{-1/2}\Gamma(\frac{1}{4})^2}{(\sqrt{2} - 1)^2\sqrt{\cos \theta}} (\frac{1}{4} + y^2)^3 e^{-\theta|y|}.$$

Monicity of the fixed degree- d polynomial h gives $|h(\frac{1}{2} + iy)| \geq c_h(1 + |y|)^d$ for some $c_h > 0$ outside a sufficiently large interval: divide its value by $(\frac{1}{2} + iy)^d$, whose quotient tends to one. On the remaining compact interval, the entire function g/h is bounded. These estimates prove finiteness of every moment of ν_h . The measure is positive away from a discrete set. Therefore the norm of a nonzero polynomial is strictly positive: zero norm would make the polynomial vanish almost everywhere on a line and hence identically.

Fix $k \geq 1$, and let

$$\begin{aligned} E^{(k)} &= E_h^{\otimes k}, & A_k &= \sum_{j=1}^k A_{h,j}, & D^{(k)} &= \sum_{j=1}^k D_j, \\ \mathcal{P}_M^{(k)} &= \mathbb{C}[t_1, \dots, t_k]_{\deg \leq M}, & \mathcal{T}^{(k)}(P) &= P(D_1, \dots, D_k) F_h^{\otimes k}, \\ \mathcal{J}(P) &= v_h^{\otimes k}[P]_{(h(t_1), \dots, h(t_k))}. \end{aligned} \tag{FD.6}$$

All tensor generators in this section are additive sums, as displayed in (FD.6). The multiplicative symmetric power of a linear map is a different construction and is not used here.

For $M \geq k(d-1)$, the map $\mathcal{J}$ on $\mathcal{P}_M^{(k)}$ is onto: each reduced monomial has each coordinate degree below d , hence total degree at most $k(d-1)$, and multiplication by $v_h^{\otimes k}$ is invertible. Its kernel and its actual theta image are

$$\mathcal{I}_M = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)), \quad \mathcal{B}_M = \mathcal{T}^{(k)}\mathcal{I}_M. \tag{FD.7}$$

Successive monic division in the variables expresses every polynomial in the kernel as $\sum_j h(t_j)P_j$. Under $\mathcal{T}^{(k)}$, each summand is an actual theta boundary in factor j , by (FD.4). Its cochain primitive has coefficient $(-1)^{j-1}$;

the tensor differential has the same coefficient, so their product gives the displayed positive summand $+h(t_j)P_j$.

Let $R_M : E^{(k)} \rightarrow \mathcal{T}^{(k)}\mathcal{P}_M^{(k)}$ be the unique least-norm lift of the full jet, using the product measure induced by (FD.5), and write $G_M = R_M^* R_M$, $\mathcal{C}_M = G_M^{-1}$. The existence, uniqueness, and inverse are also obtained directly below. The minimum is orthogonal to the specified $\mathcal{B}_M$, and its full jet is identity. Its arithmetic class is that of the original product source lift with the specified jet, since all admissible changes are the actual boundaries in (FD.7).

Start with the monic, unscaled orthogonal polynomials and their actual norms

$$p_0 = 1, \quad p_n = t^n - \text{proj}_{\mathbb{C}[t]_{\leq n-1}} t^n, \quad \omega_n = \int |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y) > 0. \quad (\text{FD.8})$$

For a multi-index $\alpha \in \mathbb{N}_0^k$, set

$$p_\alpha = \prod_{j=1}^k p_{\alpha_j}(t_j), \quad \omega_\alpha = \prod_{j=1}^k \omega_{\alpha_j}, \quad z_\alpha = \mathcal{J}(p_\alpha). \quad (\text{FD.9})$$

Let T_M have columns $\mathcal{T}^{(k)}p_\alpha$ for $|\alpha| \leq M$, let $J_M = [z_\alpha]_{|\alpha| \leq M}$, and let $\Omega_{\leq M} = \text{diag}_{|\alpha| \leq M} \omega_\alpha$. Product orthogonality gives $T_M^* T_M = \Omega_{\leq M}$. Surjectivity of J_M implies

$$\begin{aligned} \mathcal{C}_M &= J_M \Omega_{\leq M}^{-1} J_M^* = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha} > 0, \\ G_M &= \mathcal{C}_M^{-1}, \quad R_M = T_M \Omega_{\leq M}^{-1} J_M^* G_M. \end{aligned} \quad (\text{FD.10})$$

Indeed $J_M \Omega_{\leq M}^{-1} J_M^* G_M = I$. Any other coefficient vector with jet u is $\Omega_{\leq M}^{-1} J_M^* G_M u + w$, where $J_M w = 0$. Its cross term with the first vector is $w^* J_M^* G_M u = 0$, so its squared norm is exactly $u^* G_M u + w^* \Omega_{\leq M} w$. This proves the unique minimum, its orthogonality to $\mathcal{B}_M$, and $R_M^* R_M = G_M$.

1.2 The actual next relation and its recurrence bound

For $|\beta| = M + 1$, define the frontier matrices

$$\Omega = \text{diag}_{|\beta|=M+1} \omega_\beta, \quad F = [z_\beta]_{|\beta|=M+1}, \quad T_+ = [\mathcal{T}^{(k)}p_\beta]_{|\beta|=M+1}. \quad (\text{FD.11})$$

There are $r = \binom{M+k}{k-1}$ columns. They are orthogonal to the entire degree- M source image. If $P_{\mathcal{B}_M}$ denotes orthogonal projection onto the actual boundary space, put

$$\mathcal{E}_M = (1 - P_{\mathcal{B}_M})\mathcal{B}_{M+1}, \quad b_M = T_+ - R_M F : \mathbb{C}^r \longrightarrow \mathcal{E}_M. \quad (\text{FD.12})$$

The full jet of b_M is $F - F = 0$, and its two terms are orthogonal to $\mathcal{B}_M$, so its range is contained in $\mathcal{E}_M$. Its highest orthogonal polynomial component is T_+ , proving injectivity. Conversely, extract the highest component of a member of $\mathcal{E}_M$ and subtract its corresponding b_M -image. The remainder is simultaneously in $\mathcal{B}_M$ and orthogonal to $\mathcal{B}_M$, and is therefore zero. This proves that b_M is an isomorphism and constructs its inverse.

The exact Gram and cross-pairing are

$$H_M^{\text{rel}} := b_M^* b_M = \Omega + F^* G_M F, \quad b_M^* R_M = -F^* G_M. \quad (\text{FD.13})$$

The superscript distinguishes this relation Gram from the Hermitian control matrix introduced later. The projection of R_M onto the new layer and the updated representative are

$$Y_M = -b_M (H_M^{\text{rel}})^{-1} F^* G_M, \quad R_{M+1} = R_M - Y_M. \quad (\text{FD.14})$$

These corrections are actual boundaries. In particular, the full jet and the original arithmetic class stay fixed. The quotient-layer map

$$\mathcal{B}_{M+1}/\mathcal{B}_M \longrightarrow \mathcal{E}_M, \quad [z] \longmapsto (1 - P_{\mathcal{B}_M})z$$

is an isomorphism, whose inverse is the inclusion followed by quotient. Its further map into the quotient by $\mathcal{B}_{M+1}$ is zero. Write λ_M for the specified original joint support label recording the relation space $\mathcal{B}_M$, and write 0 for the zero amplitude in that support fibre. On the original supported reconstruction this is $(\lambda_M, u) \mapsto (\lambda_{M+1}, 0)$. For $u \neq 0$ the preceding class is nonzero. The final class is the supported zero of that fibre; it is distinct from τ , the original external-absence element. The notation $e = 0^\bullet$ denotes the nonabsent represented scalar zero in the original scalar reconstruction. Thus the relation Gram in (FD.13) measures the actual relations becoming represented zero at the next support.

Orthogonal projection and the finite sum in (FD.10) give

$$\begin{aligned} G_{M+1} &= G_M - G_M F (\Omega + F^* G_M F)^{-1} F^* G_M, \\ \mathcal{C}_{M+1} - \mathcal{C}_M &= F \Omega^{-1} F^*. \end{aligned} \quad (\text{FD.15})$$

The second equality also follows simply by adding the degree- $M+1$ terms to (FD.10); it does not require a limiting argument.

Set $p_{-1} = 0$. Multiplication by t on the integration line has adjoint multiplication by $1-t$. Orthogonality and monicity therefore give

$$\begin{aligned} tp_n &= p_{n+1} + b_n p_n - a_n p_{n-1}, & a_n &= \omega_n / \omega_{n-1} \quad (n \geq 1), & a_0 &= 0, \\ b_n &= \frac{\int (\frac{1}{2} + iy) |p_n(\frac{1}{2} + iy)|^2 d\nu_h(y)}{\omega_n}, & \Re b_n &= \frac{1}{2}. \end{aligned} \quad (\text{FD.16})$$

For $n \geq 1$, pairing against a polynomial of degree below $n-1$ in the expansion of tp_n vanishes by the adjoint identity. Pairing against p_{n-1} gives $-\omega_n/\omega_{n-1}$, and pairing against p_n gives the displayed b_n . For $n = 0$, monicity gives $tp_0 = p_1 + b_0 p_0$, and the term $a_0 p_{-1} = 0$ is exactly the declared zero term. This proves every coefficient in the recurrence. The minus sign and the possibly nonzero imaginary part of b_n are retained.

Define

$$w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, \quad E_+ = [w_\beta]_{|\beta|=M+1}. \quad (\text{FD.17})$$

Here e_j is the standard multi-index vector; it is not the supported scalar $e = 0^\bullet$. The full arithmetic jet is a module map, so $\mathcal{J}((\sum_j t_j)P) = A_k \mathcal{J}(P)$. Inserting (FD.16) into (FD.10) proves

$$\mathcal{Z}_M := A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*. \quad (\text{FD.18})$$

To verify the cancellation explicitly, for every internal edge $\alpha \rightarrow \alpha + e_j$, the raising term has coefficient $1/\omega_\alpha$, while the lowering term has coefficient $-a_{\alpha_j+1}/\omega_{\alpha+e_j} = -1/\omega_\alpha$. Their adjoints cancel as well. The diagonal contribution is $2 \sum_j \Re b_{\alpha_j} = k$. The only uncanceled edges have $|\alpha| = M$ and $|\alpha + e_j| = M+1$, and their coefficients are exactly those in the right side of (FD.18).

For completeness, the arithmetic derivative defect has the same actual relation coordinates. Set

$$B_M^{\text{der}} = D^{(k)} R_M - R_M A_k, \quad C_M^{\text{rel}} = (1 - P_{\mathcal{B}_M}) B_M^{\text{der}}.$$

Its full jet is zero, and its highest polynomial component is $T_+ \Omega^{-1} E_+^* G_M$. The latter assertion follows from the coefficient of p_β in $(\sum_j t_j) R_M$: it is $\sum_{j: \beta_j > 0} z_{\beta - e_j}^* G_M / \omega_{\beta - e_j} = w_\beta^* G_M / \omega_\beta$. The layer isomorphism (FD.12)

therefore proves

$$\begin{aligned}
C_M^{\text{rel}} &= b_M \Omega^{-1} E_+^* G_M, \\
W_M &:= A_k^* G_M + G_M A_k - k G_M \\
&= G_M \mathcal{Z}_M G_M \\
&= -(Y_M^* C_M^{\text{rel}} + (C_M^{\text{rel}})^* Y_M).
\end{aligned} \tag{FD.19}$$

The last equality follows by substituting (FD.13) and (FD.14); in particular its minus sign is retained.

The recurrence constant is

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}) > 0. \tag{FD.20}$$

Strict positivity follows from $a_{n+1} > 0$ for every $n \geq 0$. We now prove its complete lowering estimate.

For coefficient calculations only, use the exact coordinate map that divides the basis vector p_α by $\sqrt{\omega_\alpha}$. Its inverse multiplies by $\sqrt{\omega_\alpha}$; the original polynomials, their norms, and the measure are unchanged. The degree- (-1) coefficient space is by definition $\{0\}$; in particular every lowering map from degree zero has that zero codomain. This retains the admitted endpoint $d = 1, M = 0$. In these coefficient spaces let R_j be the raising map $e_\alpha \mapsto \sqrt{a_{\alpha_j+1}} e_{\alpha+e_j}$, and let R_j^* be its lowering adjoint. Put

$$Q_M^{\text{jet}} = [z_\alpha / \sqrt{\omega_\alpha}]_{|\alpha|=M}, \quad B^{\text{inc}} = (\sum_j R_j)^*|_{M+1 \rightarrow M}.$$

The retained coefficients in (FD.17) give exactly $E_+ \Omega^{-1/2} = Q_M^{\text{jet}} B^{\text{inc}}$. Raising and lowering in distinct coordinates commute. Hence, on the degree- M coefficient space,

$$B^{\text{inc}} (B^{\text{inc}})^* = \text{diag}_{|\alpha|=M} \sum_j (a_{\alpha_j+1} - a_{\alpha_j}) + L^* L, \quad L = \sum_j R_j^*|_{M \rightarrow M-1}.$$

For every coefficient vector c , Cauchy–Schwarz gives $\|Lc\|^2 \leq k \sum_j \|R_j^* c\|^2$. Thus

$$L^* L \preceq k \text{diag}_{|\alpha|=M} \sum_j a_{\alpha_j}, \quad B^{\text{inc}} (B^{\text{inc}})^* \preceq \Gamma_{k,M} I.$$

Multiplication by Q_M^{jet} and its adjoint, followed by inclusion of the positive degree- M sum in (FD.10), proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Q_M^{\text{jet}} (Q_M^{\text{jet}})^* \preceq \Gamma_{k,M} \mathcal{C}_M. \tag{FD.21}$$

1.3 Exact symmetric restriction with every orbit factor

Let P_π be ordinary factor permutation. The relevant module is

$$E_k^{\text{sym}} = (E_h^{\otimes k})^{S_k}, \quad N = \dim E_k^{\text{sym}} = \binom{k+d-1}{d-1}.$$

It is the actual top-degree summand selected on the original cochain tensor product by $S_k = (1/k!) \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi$, where T_π is permutation with its Koszul sign. In top degree $T_\pi = \text{sgn}(\pi) P_\pi$, so the two signs cancel. The numerator squares to $k!$ times itself, proving the idempotent with its literal coefficient $1/k!$. In particular, every repeated tensor $v^{\otimes k}$ remains in this summand.

Use the quotient basis formed by the unscaled orbit sums of $[t^{a_1}] \otimes \cdots \otimes [t^{a_k}]$, $0 \leq a_i < d$. Let I_k be its coefficient inclusion in the full tensor basis. If n_j counts factors with exponent j , its orbit size is $k! / \prod_{j=0}^{d-1} n_j!$. The orbit supports are disjoint, so

$$D_k = I_k^* I_k = \text{diag} \left(\frac{k!}{\prod_{j=0}^{d-1} n_j!} \right), \quad L_k = D_k^{-1} I_k^*, \quad L_k I_k = I. \quad (\text{FD.22})$$

The projector onto invariant coefficient vectors is $\Pi_k = I_k L_k$. Permutations preserve the source, the relation ideal, the product measure, and $v_h^{\otimes k}$. Uniqueness of the minimum therefore gives equivariance of R_M , and consequently of G_M and $\mathcal{C}_M$. The exact restricted matrices are

$$G_M^{\text{sym}} = I_k^* G_M I_k, \quad \mathcal{C}_M^{\text{sym}} = L_k \mathcal{C}_M L_k^*, \quad A_k I_k = I_k A_k^{\text{sym}}. \quad (\text{FD.23})$$

The first two matrices are inverse. For example,

$$G_M^{\text{sym}} \mathcal{C}_M^{\text{sym}} = I_k^* G_M \Pi_k \mathcal{C}_M L_k^* = I_k^* \Pi_k L_k^* = I,$$

because both G_M and $\mathcal{C}_M$ commute with Π_k , their product is identity, $\Pi_k L_k^* = L_k^*$, and $I_k^* L_k^* = (L_k I_k)^* = I$.

In the source frontier let Q_+ be the coefficient inclusion matrix whose columns are the 0/1 orbit indicators in the ordered basis $(p_\beta)_{|\beta|=M+1}$. Thus each column represents the unscaled orbit sum of those p_β . If $m_j(\beta)$ counts the entries of β equal to j , the corresponding source orbit norm is exactly

$$\omega_{[\beta]} = \frac{k!}{\prod_{j \geq 0} m_j(\beta)!} \prod_{i=1}^k \omega_{\beta_i}. \quad (\text{FD.24})$$

This is a sum of equal squared norms of pairwise orthogonal distinct product polynomials. Define the actual restricted frontier matrices

$$\Omega_s = Q_+^* \Omega Q_+, \quad F_s = L_k F Q_+, \quad E_s = L_k E_+ Q_+. \quad (\text{FD.25})$$

Thus Ω_s is the positive diagonal matrix with entries (FD.24); none of its orbit factors are suppressed.

The source invariant projector is $\Pi_+ = Q_+(Q_+^* Q_+)^{-1} Q_+^*$. Since Ω is constant on each orbit, it commutes with Π_+ , and direct calculation in one orbit gives

$$Q_+ \Omega_s^{-1} Q_+^* = \Omega^{-1} \Pi_+.$$

Indeed, on an orbit of size m and weight ω , the left side is the all-ones matrix divided by $m\omega$; this is precisely $\omega^{-1} \Pi_+$ there. Equivariance implies $L_k F = L_k F \Pi_+$ and $L_k E_+ = L_k E_+ \Pi_+$: averaging any source vector before applying an equivariant map has the same invariant component as applying the map first. Congruence of (FD.15), (FD.18), and (FD.21) therefore gives, with

$$\begin{aligned} \mathcal{C} &= \mathcal{C}_M^{\text{sym}}, \quad G = G_M^{\text{sym}} = \mathcal{C}^{-1}, \quad A = A_k^{\text{sym}}, \\ \Delta \mathcal{C} &= \mathcal{C}_{M+1}^{\text{sym}} - \mathcal{C}_M^{\text{sym}}, \quad \mathcal{Z} = A\mathcal{C} + \mathcal{C}A^* - k\mathcal{C}, \end{aligned} \quad (\text{FD.26})$$

the exact formulas

$$\begin{aligned} \Delta \mathcal{C} &= F_s \Omega_s^{-1} F_s^*, \\ \mathcal{Z} &= F_s \Omega_s^{-1} E_s^* + E_s \Omega_s^{-1} F_s^*, \\ E_s \Omega_s^{-1} E_s^* &\preceq \Gamma_{k,M} \mathcal{C}. \end{aligned} \quad (\text{FD.27})$$

For the operator term, equivariance also gives $L_k A_k = A_k^{\text{sym}} L_k$, proving that the congruence of $A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k\mathcal{C}_M$ is exactly the $\mathcal{Z}$ in (FD.26).

1.4 The aggregate spectral-position restriction

Positive square roots below are the unique positive square roots of the retained positive matrices. They are used as explicit comparison maps; their inverses are retained, and neither the source measure nor the arithmetic metric is redefined. Set

$$\begin{aligned} X &= G^{1/2} F_s \Omega_s^{-1/2}, \quad Y = G^{1/2} E_s \Omega_s^{-1/2}, \\ K &= X X^* = G^{1/2} \Delta \mathcal{C} G^{1/2}, \quad H = G^{1/2} \mathcal{Z} G^{1/2} = X Y^* + Y X^*. \end{aligned} \quad (\text{FD.28})$$

The last inequality of (FD.27) gives $Y Y^* \preceq \Gamma_{k,M} I$, and hence $Y^* Y \preceq \Gamma_{k,M} I$ on its own coefficient space. Here the latter statement follows from $\|Y\|_{\text{op}}^2 =$

$\|YY^*\|_{\text{op}}$, or directly from the singular values. For the Frobenius norm $\|T\|_F^2 = \text{tr}(T^*T)$,

$$\|XY^*\|_F^2 = \text{tr}(X^*X Y^*Y) \leq \Gamma_{k,M} \text{tr}(X^*X).$$

The norm of a matrix equals the norm of its adjoint, so the triangle inequality gives the proved upper estimate

$$\|H\|_F^2 \leq 4\Gamma_{k,M} \text{tr} K. \quad (\text{FD.29})$$

List the roots of h , repeated with their original algebraic multiplicities, as $\rho_1, \dots, \rho_d$, and put

$$\delta_i = \Re \rho_i - \frac{1}{2}, \quad \eta_{\mathbf{n}} = \sum_{i=1}^d n_i \rho_i, \quad \mathbf{n} \in \mathbb{N}_0^d, \quad |\mathbf{n}| = k. \quad (\text{FD.30})$$

The eigenvalues of A , with algebraic multiplicity, are exactly the $\eta_{\mathbf{n}}$. To prove this while retaining nilpotents, triangularize A_h by induction using an eigenvector and the induced operator on the quotient. In a basis with $A_h e_j = \rho_j e_j + \sum_{i < j} c_{ij} e_i$, applying the additive generator to a symmetric monomial of occupations $\mathbf{n}$ gives diagonal coefficient $\sum_j n_j \rho_j$. Every other term replaces an index j by $i < j$, reducing the sum of the indices. Ordering monomials by that sum, and resolving ties arbitrarily, makes the operator triangular with precisely the displayed diagonal. Each weak composition contributes once; equal eigenvalues remain repeated with their algebraic multiplicities. The triangular off-diagonal entries are retained, including those caused by nontrivial Jordan blocks.

Let $B = G^{1/2} A G^{-1/2}$. Substitution into (FD.26) gives $H = B + B^* - kI$. Unitary Schur triangularization gives $T = U^* B U$, upper triangular with diagonal entries $\eta_{\mathbf{n}}$. Such a triangularization is obtained inductively by taking a unit eigenvector as the first basis vector and applying the induction to the compression on its orthogonal complement. Unitary conjugation preserves the Frobenius norm. Its diagonal and off-diagonal entries give the exact identity

$$\|H\|_F^2 = \sum_{|\mathbf{n}|=k} (2\Re \eta_{\mathbf{n}} - k)^2 + 2 \sum_{i < j} |T_{ij}|^2 \geq 4 \sum_{|\mathbf{n}|=k} \left(\sum_i n_i \delta_i \right)^2. \quad (\text{FD.31})$$

For $i < j$, the two corresponding entries of $T + T^* - kI$ are T_{ij} and $\overline{T_{ij}}$; this proves the factor two. Thus the retained nilpotent and nonnormal contributions cannot weaken the displayed lower bound.

We evaluate its sum exactly. For $d \geq 2$, the ordinary generating functions for weak compositions give

$$\begin{aligned}\sum_{|\mathbf{n}|=k} n_1 &= \binom{k+d-1}{d}, \\ \sum_{|\mathbf{n}|=k} n_1(n_1-1) &= 2 \binom{k+d-1}{d+1}, \\ \sum_{|\mathbf{n}|=k} n_1 n_2 &= \binom{k+d-1}{d+1}.\end{aligned}\tag{FD.32}$$

Indeed their generating functions are respectively

$$z(1-z)^{-d-1}, \quad 2z^2(1-z)^{-d-2}, \quad z^2(1-z)^{-d-2}.$$

These follow by differentiating the appropriate geometric-series factors. Extracting the coefficient of z^k gives (FD.32). Write $a = \binom{k+d-1}{d}$ and $b = \binom{k+d-1}{d+1}$. Then the square expands as

$$(a+2b) \sum_i \delta_i^2 + 2b \sum_{i<j} \delta_i \delta_j = (a+b) \sum_i \delta_i^2 + b \left(\sum_i \delta_i \right)^2.$$

Pascal's identity therefore proves

$$\begin{aligned}\mathcal{S}_{h,k} &:= \sum_{|\mathbf{n}|=k} \left(\sum_i n_i \delta_i \right)^2 \\ &= \binom{k+d}{d+1} \sum_i \delta_i^2 + \binom{k+d-1}{d+1} \left(\sum_i \delta_i \right)^2.\end{aligned}\tag{FD.33}$$

Binomial coefficients with smaller upper than nonnegative lower index are zero. For $d=1$, direct evaluation gives $k^2 \delta_1^2$; the right side of (FD.33) is $[\binom{k+1}{2} + \binom{k}{2}] \delta_1^2 = k^2 \delta_1^2$, so the same formula holds in that case.

Combining (FD.29), (FD.31), and (FD.33) proves the aggregate arithmetic restriction

$$\boxed{\Gamma_{k,M} \operatorname{tr} (G_M^{\operatorname{sym}} (\mathcal{C}_{M+1}^{\operatorname{sym}} - \mathcal{C}_M^{\operatorname{sym}})) \geq \mathcal{S}_{h,k}.}\tag{FD.34}$$

The trace in this formula is exactly $\operatorname{tr} K$, by cyclicity. All its matrices were defined from the actual measure, complete arithmetic jet unit, and unscaled orbit norms.

1.5 Coordinate invariance and determinant growth

The comparison maps have an exact coordinate-invariance statement. For invertible changes S in quotient coordinates and R in frontier coordinates, set

$$\begin{aligned} A' &= SAS^{-1}, & \mathcal{C}' &= SCS^*, & G' &= S^{-*}GS^{-1}, \\ F'_s &= SF_sR, & E'_s &= SE_sR, & \Omega'_s &= R^*\Omega_sR. \end{aligned} \quad (\text{FD.35})$$

Here $S^{-*} = (S^{-1})^*$. Cancellation of R, R^{-1} in the factorizations proves $\Delta\mathcal{C}' = S\Delta\mathcal{C}S^*$, and substitution in the generator identity gives $\mathcal{Z}' = S\mathcal{Z}S^*$. The lowering inequality is carried through the same congruence. Direct cyclic cancellation and determinant multiplicativity give

$$\begin{aligned} \text{tr}(G'\Delta\mathcal{C}') &= \text{tr}(G\Delta\mathcal{C}), \\ \frac{\det(\mathcal{C}' + \Delta\mathcal{C}')}{\det\mathcal{C}'} &= \frac{\det(\mathcal{C} + \Delta\mathcal{C})}{\det\mathcal{C}}. \end{aligned} \quad (\text{FD.36})$$

Furthermore $U_S = (\mathcal{C}')^{-1/2}S\mathcal{C}^{1/2}$ satisfies $U_S U_S^* = I$, and hence is unitary, with $H' = U_S H U_S^*$. This follows by substituting $\mathcal{C}' = SCS^*$ and $\mathcal{Z}' = S\mathcal{Z}S^*$. Thus the exact trace, determinant ratio, and Frobenius quantity are preserved by the comparison-coordinate maps; no arithmetic normalization has been imposed.

The exact update and (FD.28) give

$$\frac{\det\mathcal{C}_{M+1}^{\text{sym}}}{\det\mathcal{C}_M^{\text{sym}}} = \det(I + K). \quad (\text{FD.37})$$

Indeed congruence by $\mathcal{C}^{-1/2} = G^{1/2}$ takes $\mathcal{C} + \Delta\mathcal{C}$ to $I + K$, and its determinant multiplier is $1/\det\mathcal{C}$. If $\kappa_1, \dots, \kappa_N \geq 0$ are the eigenvalues of K , expanding $\prod_j (1 + \kappa_j)$ gives $\det(I + K) \geq 1 + \sum_j \kappa_j = 1 + \text{tr} K$. Since $\Gamma_{k,M} > 0$ was proved in (FD.20), (FD.34) now proves

$$\boxed{\frac{\det\mathcal{C}_{M+1}^{\text{sym}}}{\det\mathcal{C}_M^{\text{sym}}} \geq 1 + \frac{\mathcal{S}_{h,k}}{\Gamma_{k,M}}.} \quad (\text{FD.38})$$

For any integers $M_1 > M_0 \geq k(d-1)$, multiplying the finitely many positive inequalities (FD.38) telescopes to

$$\boxed{\frac{\det\mathcal{C}_{M_1}^{\text{sym}}}{\det\mathcal{C}_{M_0}^{\text{sym}}} \geq \prod_{M=M_0}^{M_1-1} \left(1 + \frac{\mathcal{S}_{h,k}}{\Gamma_{k,M}} \right).} \quad (\text{FD.39})$$

This is a finite product over the specified actual support degrees. It requires no limit or unproved asymptotic estimate.

For a reflection-stable packet, the original reflection maps each zero ρ to $1 - \bar{\rho}$, retaining its full order. Each paired displacement δ_i is sent to $-\delta_i$, so $\sum_i \delta_i = 0$. Hence the exact coefficient becomes

$$\boxed{\mathcal{S}_{h,k} = \binom{k+d}{d+1} \sum_{i=1}^d (\Re \rho_i - \tfrac{1}{2})^2.} \quad (\text{FD.40})$$

In particular, the presence of any actual off-axis zero makes this coefficient strictly positive. It forces a positive relative frontier increment and the determinant lower bound at every admitted support degree, within the actual symmetric summand.

1.6 The exact signed eigenline witness

Let $v \neq 0$ be an actual eigenvector of A_h , with $A_h v = \rho v$. In the unscaled symmetric quotient coordinates put $u = L_k(v^{\otimes k})$, so $I_k u = v^{\otimes k}$. This vector is nonzero: choose a linear functional ℓ with $\ell v = 1$; then $\ell^{\otimes k}(v^{\otimes k}) = 1$. The additive generator satisfies $Au = k\rho u$. With the exact positive number $q = u^*Gu$, put

$$x = \Omega_s^{-1/2} F_s^* Gu, \quad y = \Omega_s^{-1/2} E_s^* Gu. \quad (\text{FD.41})$$

The symmetric restriction of (FD.19) is $W = A^*G + GA - kG = GZG$. On the eigenvector,

$$u^*Wu = k(2\Re \rho - 1)q.$$

On the other hand, (FD.27) and (FD.41) give $u^*GZGu = x^*y + y^*x = 2\Re(x^*y)$. Equating these two exact expressions proves the signed identity

$$\boxed{\Re(x^*y) = k(\Re \rho - \tfrac{1}{2})q.} \quad (\text{FD.42})$$

Thus this correlation has the retained sign of the actual real-part displacement. No absolute-value estimate was used in obtaining it.

Define the two directional costs and the actual symmetric maximum cost

$$\alpha = \frac{\|x\|^2}{q}, \quad \beta = \frac{\|y\|^2}{q}, \quad \lambda_{k,M}^{\text{sym}} = \max_{w \neq 0} \frac{w^* \Delta \mathcal{C} w}{w^* \mathcal{C} w} = \lambda_{\max}(K). \quad (\text{FD.43})$$

The last equality follows by the invertible substitution $w = \mathcal{C}^{-1/2}z$ in the Rayleigh quotient. The lowering estimate in (FD.27), evaluated at Gu ,

gives $\beta \leq \Gamma_{k,M}$, because $(Gu)^*\mathcal{C}(Gu) = q$. The same substitution into the maximum defining $\lambda_{k,M}^{\text{sym}}$ gives $\alpha \leq \lambda_{k,M}^{\text{sym}}$. Cauchy–Schwarz applied to (FD.42) gives

$$\alpha\beta \geq k^2(\Re\rho - \tfrac{1}{2})^2.$$

Consequently the actual arithmetic costs satisfy

$$\boxed{\Gamma_{k,M}\lambda_{k,M}^{\text{sym}} \geq k^2(\Re\rho - \tfrac{1}{2})^2.} \tag{FD.44}$$

Every term in the signed identity and this bound is attached to the actual next original relation through (FD.12), (FD.19), and the exact symmetric restrictions.

Exact scope of the resulting restriction. Equations (FD.34)–(FD.40) and (FD.42)–(FD.44) prove the stated frontier and determinant restrictions for the original arithmetic family, including all multiplicities and nilpotent actions. The source explicitly records the unestimated quantity: A1829, line 5580, states that no uniform sublinear excess was established; A1859, line 6013, states that the finite bound’s constants were not bounded sublinearly in tensor degree; A1859, line 6327, states that the uniform sublinear estimate for the actual arithmetic family was not proved. No upper estimate incompatible with (FD.38) or (FD.44) is asserted here. These are proved necessary restrictions on an actual RH counterexample, not an assertion that their arithmetic values cannot satisfy them.